3 Future Work and Recommendations

We outline several actionable recommendations for the Information Retrieval (IR) community, emphasizing practical steps for researchers, conference organizers, journal editors, and institutions. While these recommendations are primarily targeted at IR, they also reflect broader best practices in academia.

3.1 Researchers should familiarize themselves with research-assistive AI tools

At present, tools like the AI Scientist hold little practical value. In contrast, more general AI tools, such as ChatGPT for text generation and GitHub Copilot for programming, already provide substantial benefits in research, improving writing, coding, and brainstorming efficiency. Researchers who ignore these readily available AI-powered assistants are already at a disadvantage compared to their peers who integrate them into their workflows.

Looking ahead, tools like the AI Scientist will become vastly more powerful and their use may be ubiquitous. Unlike today's prototypes, we forecast that future iterations will reliably generate research ideas, execute complex experiments, and draft high-quality manuscripts with minimal human oversight. Notably, such research-assistive AI capitalize on the underlying emergent capabilities and rapid performance progress of their underlying foundation models.

Early-career researchers, in particular, must closely monitor these developments. The use of AI in research will soon become indispensable. Those who fail to adopt AI tools risk falling behind their peers, akin to a researcher today relying on a typewriter and calculator instead of adopting a computer with text editor and programming language support. The future of research belongs to those who embrace AI-enhanced workflows and integrate them into their daily scientific practice.

The academic community must prepare for this shift and ideally lead the discussion on how AI should be integrated into scientific research. To do so, researchers should actively engage with tools like the AI Scientist, follow advancements in this domain, and, where possible, contribute to their development and define their appropriate use. By understanding and shaping these tools, researchers ensure that AI development aligns with rigorous scientific principles rather than being dictated solely by industry interests.

3.2 Updated Guidelines for Authors and Researchers on the Use of AI

The ACM and other publishers provide guidelines on using generative AI in manuscript preparation. Currently, ACM permits AI to assist in manuscript writing, including generating text, images, tables, and code[21]. However, the potential for AI to conduct research and autonomously generate entire manuscripts and experimental workflows has not yet been addressed by these policies.

We urge the IR community and publishers, such as ACM, to initiate discussions on how AI-driven research tools, like the AI Scientist, should best be integrated into scientific workflows. Specifically, the community should establish policies as to what extent AI-generated research can be submitted to conferences such as SIGIR.

At present, our assessment suggests that tools like the AI Scientist are not yet reliable enough to contribute meaningfully to research. Consequently, manuscripts generated entirely by AI should not be accepted for publication at this time. However, this stance will need to be revisited regularly as AI capabilities evolve rapidly.

We recommend that the academic community, particularly editors of journals and members of steering or ethics committees, take an active role in revising guidelines regarding the use of AI tools in research. These revisions should be developed in collaboration with experts who possess in-depth knowledge of the capabilities and limitations of

[21] https://www.acm.org/publications/policies/frequently-asked-questions#h-can-i-use-generative-ai-software-tools-to-prepare-my-manuscript

AI systems, such as the AI Scientist. Additionally, input from stakeholders like peer reviewers, program chairs of conferences, and other members of the academic publishing ecosystem will be essential to ensure that these tools are integrated effectively while maintaining academic integrity and quality control.

3.3 Addressing the Risks of AI-Generated Assignments and Homeworks

The ability of AI tools such as the AI Scientist to generate well-structured academic papers that are hard to distinguish from student submissions poses an immediate challenge for universities. These systems can produce coherent, plausible reports that, especially under time constraints, may not be easily recognized as AI-generated by instructors. This presents a fundamental risk to academic integrity, as students can now generate entire assignments without engaging in the learning process. The issue is no longer speculative—AI-generated work is already at a level where instructors may fail to detect it without in-depth scrutiny. It is also difficult to prosecute, as the stochastic nature of such tools produce diverse output that is difficult to attribute back to a plagiarised source. Universities must therefore act urgently to adapt their assessment methodologies and reinforce academic standards.

A first step is to rethink assignment design, prioritizing evaluations that require learners to demonstrate personal mastery of the subject matter. This can be through iterative work, personal reflection, or in-person verification. Traditional take-home essays should be supplemented with oral examinations, in-class assignments, and project-based assessments where students demonstrate an evolving understanding of the subject, which adds value beyond what AI tools yield as a first draft. Additionally, universities should establish clear policies on AI usage, distinguishing acceptable from unacceptable applications of these tools and incorporating AI literacy into curricula. Educating students on the ethical implications and potential academic consequences of AI misuse can encourage responsible engagement. Furthermore, while AI-detection tools remain unreliable, institutions should implement verification mechanisms such as requiring students to submit intermediate drafts, video explanations, or interactive discussions about their work. These measures, when combined with updated academic integrity policies and structured instructor training, can help mitigate the risks posed by AI-generated assignments and uphold the value of university education.

3.4 Benchmarking AI-Driven Research Agents in Information Retrieval

We propose developing structured benchmarks to evaluate AI-driven research agents, recognizing the need for broader community discussion to establish consensus on assessment criteria. Such benchmarks are essential to determine when AI agents achieve "Scientific AGI," the point at which they can autonomously conduct research at human-level quality.

A critical area is AI-powered peer review. These systems generate structured reviews, yet their reliability remains untested. Benchmarks could include expert annotations, inter-reviewer agreement analyses, and alignment with evaluation criteria to compare AI and human reviews.

AI-driven idea generation also requires assessment. Evaluations could compare AI-generated ideas with expert proposals, analyzing acceptance rates and citation impact. Experimental AI agents should be benchmarked for their ability to implement methodologies, configure parameters, and ensure reproducibility. Failure analysis is crucial to detect methodological flaws.

Manuscript generation benchmarks should go beyond fluency, assessing argument structuring, literature integration, and empirical accuracy. Automated inconsistency detection, including hallucinated citations, is essential. Blinded peer review comparisons and authorship attribution frameworks may help distinguish AI contributions.

Other AI-assisted tasks, including literature review and dataset selection, require evaluation. AI benchmarks could include retrieval relevance metrics, citation analysis, and bias detection to meet expert standards.

Developing a comprehensive benchmarking suite requires collaboration across academia, industry, and publishers. Standardized datasets, evaluation protocols, and performance metrics would help ensure scientific rigor and meaningful AI integration into research. Establishing these benchmarks will provide clear indicators of AI progress toward autonomous, human-level scientific inquiry.

3.5 Release of AI Research Logs for Transparency

A key advantage of AI-generated research is its ability to systematically log every step of the process. Unlike human researchers, AI systems can capture all experimental iterations, hyperparameter adjustments, and decision rationales in real time. To maximize transparency and reproducibility, AI-generated studies should include execution logs detailing raw input prompts, intermediate code modifications, and abandoned experimental runs. These logs would enable human and AI researchers to trace the AI's reasoning process, identify potential errors, and refine methodologies.

Comprehensive versioning of all research artifacts should be mandatory. Every change to code and experimental setups should be stored in structured repositories, such as Git with detailed commit histories. Timestamped metadata should accompany these changes, documenting the AI model, system configuration, and key decision points. This will allow researchers to track the evolution of AI-generated research and compare different experimental approaches systematically.

To further improve reproducibility, AI-generated research should be accompanied by containerized environments encapsulating dependencies and configurations. Using tools like Docker or Singularity, researchers should provide prebuilt containers that replicate the AI's computational environment. Additionally, automated reports summarizing why specific methodologies were selected or discarded would provide insight into the AI's decision-making process, helping both human researchers and future AI agents build on existing work more effectively.

3.6 Pilot Projects and Competitions

The IR community can learn from initiatives such as ICLR 2025 and NeurIPS, which have explored generative AI's role in peer review and manuscript submission through pilot projects. Expanding on these efforts, targeted pilot projects can further investigate AI's impact on scientific research.

One promising initiative is a competition assessing AI-generated research. We propose integrating such a competition into a major conference like SIGIR, where AI-generated papers would be evaluated alongside human-authored research through a blind peer-review process. This would provide an unbiased comparison of AI's ability to conduct rigorous scientific inquiry.

While current AI research tools remain limited, a competition of this nature could become viable within the next few years. Early planning and discussion within the community would ensure that appropriate evaluation frameworks and ethical considerations are established ahead of time, enabling a structured and fair assessment of AI-driven scientific contributions.

3.7 AI-Generated Content Tagging: Research Attribution Markup Language (RAML)

As AI-generated research becomes more prevalent, distinguishing between human-authored and AI-assisted content is critical for transparency, academic integrity, and reproducibility. Existing solutions for image watermarking and document metadata do not adequately address the need for systematic attribution in text-based research outputs. Therefore, we propose the Research Attribution Markup Language (RAML), a standardized method to tag AI-generated content within academic manuscripts.